HEC Paris MASTER IN INTERNATIONAL FINANCE

HEC Paris MAJOR IN INTERNATIONAL FINANCE
OF THE MASTER IN MANAGEMENT

X-HEC DD MASTER & MAJOR IN DATA & FINANCE

RESEARCH PAPER – TOPIC PROPOSALS

2025 - 2026 Academic Year

A Word from the Academic Directors

Dear Students,

Per requirements of your master's degree at HEC Paris, you are required to write a Research Paper (RP, henceforth). The RP accounts for 20 ECTS credits for the Master in International Finance (MIF), Major in International Finance (MIF) of the Master in Management (MiM), as well Master/Major in Data & Finance (MD&F) X-HEC Double-Degree students.[1] These 20 ECTS reflect the importance of the RP for your academic learning at HEC Paris, as they account 1/3rd to 1/6th of your diploma's total ECTS requirements depending on the program you are in.

In the case of differences between this document and the RP Guidelines provided by HEC Paris, this document prevails (given the leeway accorded to academic directors of individual master programs).

We suggest that you take your time when browsing through the pages of research topics proposed by the research faculty of HEC Paris and other lecturers teaching in the program. As you do so, please kindly note the following important points:

a. Research papers can only be written by pairs of students. Put differently, all research papers must have two students as authors (not more, not less). You can pair up with anybody from your cohort (irrespective of the program) given that the RP requirements are exactly the same across the programs indicated above.

b. Please note that the submission and defense dates are as follows:

   Submission: **before** 12:00 pm (noon) on **Monday June 1st, 2026**
   Defense: between **June 15 and June 30, 2026**

   Note that the RP submission and defense can be done earlier (say April 2026) **if** your co-author, advisor and the faculty-member who serves as defense jury's president (see below) would agree to it. Note also that faculty who will be judging your defense need to be given two weeks to read your RP.

c. RPs cannot be supervised by advisors whose names are not in this list of topics and who aren't pre-approved by the academic directors. Unapproved RP directors will be asked to rescind their supervision if prior approval hasn't been sought by the students and given by the Academic Directors (who can say "no" to proposed names): before contacting a possible RP director whose name does not appear on this document make sure that you consult Prof. Örs (or run the risk of having to explain to that person that you have wasted her/his time for nothing).

[1] As a reminder: MIF students need to validate 120 ECTS credit-hours to obtain their degree, MD&F students need 70 ECTS during the M2-year at HEC Paris (X-HEC DD in D&F degree requires 140 ECTS over two years), and MiM students 60 ECTS while in their M2-year (HEC Paris MiM degree requires 180 or 120 ECTS over three or two years depending on entry during L3 or M1, respectively).

d. Deadline for choosing a subject and agreeing with an advisor is 12:00 pm (noon) on Friday November 14th, 2025 (i.e., you have almost 1.5 months to exchange with faculty members, settle on a topic agreed by a potential RP advisor). Please note that you need to be to the point in selecting a topic and an advisor: our observation is that too much time is wasted in "shopping around": be courteous to the faculty members you spoke with and keep them informed of your final decision. Also note that Prof. Y won't be more lenient than Prof. X, as the more or less the same standards will apply when grading the RPs.

e. All RPs will have to be presented to another HEC Paris faculty or MIF lecturer during a 30-minutes RP-defense that can be in-person or online. The jury president would have read your written RP beforehand. That is, you will have to argument as to why your RP topic is interesting, explain the essence of your work, convey the inherent quality of your work, and the relevance of your findings to a qualified person other than the one who supervised your work. Both the supervisor and the person who acts as the jury in your defense will be providing points (out of 10, each) that will go towards your overall RP grade (out of 20).

f. You can choose among the many research topics proposed below by our colleagues or propose your own topic: as of today, there are more RP ideas than the number of student pairs if you consider the fact that many of the proposed subjects are open-ended.

g. Any HEC Paris faculty member can only take on a maximum of 10 (ten) research paper supervisions, whereas any MIF lecturer can only take on a maximum of 5 (five) RPs. As a result, the topics are available on a first-come-first-served basis (though, same topics can be examined with different data or different angles by different student pairs under the same advisor). RP supervisors have been warned that advisors who allow a single-authored research paper or take on more than their maximum limit (of 10 or 5 depending on their status) will be over-ruled by Prof. Örs, who is the senior academic director of your programs.

h. Your RP should be 30 pages of **main text** excluding the title page, the abstract page, table of contents (which should be prior to the main text), and the references, figures, tables, and appendixes if any (which should be after the main text). Please note that top scientific journals in economics and finance (which publish papers that make the largest contributions to these fields) require this format. Please no **not** insert tables and figures into the main text, which should be text only.

i. You should create a DropBox folder with various folders in which all RP related materials should be deposited. This DropBox folder should be accessible to your advisor, RP defense jury president, Prof. Alina Roşu and Prof. Evren Örs. You should make sure that your referenced papers (i.e., papers you cite in your literature review), your data files, your program files, etc., are in their appropriately named folder in that DropBox. The naming convention of the DropBox folder

should be: MIF2025_AdvisorLastName_StudentLastName1_StudentLastName2. **Having such a RP DropBox folder is a must.**

j. HEC Paris requires that all written work be submitted for plagiarism checks, which could include this year checks for the use of AI-driven online writing tools.

   Regarding AI & writing: good writing skills are essential for your career to convey your thoughts and contributions in various formats (email, short or long memos, longer reports). Any such skill requires exercise. Moreover, a text written by another (be it another person or AI-driven tool) is harder to revisit and improve upon than a text that directly conveys your own thoughts with your own words. You are sent a separate document that details how we expect you to do your research paper writing to avoid suspicions of plagiarism.

k. All RP advisors are required to grade the research papers based on the same high standards. Before being asked to submit their research paper topics, the supervisors (who will also act as jury for other RPs) were clearly reminded that HEC Paris is an institution with the highest academic standards conferring highly-valued academic degrees. As a result, at HEC Paris, within MIF, MiM's MIF and MD&F, an "A" stands for "Excellent" (i.e., "outstanding") work, a "B" for "Very Good" output (which means what it says), a "C" for "Good" research paper, etc.

   If the student-pair's output is truly excellent, they should expect an "A"; if it is not, then they shouldn't be surprised to get a "B", a "C" or a worse grade. The advisors and the defense jury presidents are highly qualified academics or industry professionals who have the required expertise and experience to be the sole judges of the quality of the RP papers. Be advised that if the student pair's work is very unevenly distributed (i.e., one appears to have contributed much more than the other), the same RP can lead to two different grades (say, an A and a B, or a B and a C, etc)!

l. Each supervisor can only suggest one research paper for a prize: there is always a first, even among equals. The fact that your research paper received the letter grade of "A" does not automatically qualify it for submission for a prize by any means.

m. Last but not the least, given that you have all taken either the Empirical Methods in Corporate Finance or the the Empirical Methods in Capital Markets and given that this is a Research Paper exercise, to merit an "A" grade each RP should ideally involve:

   one or more clearly motivated (based on existing research) and clearly defined hypotheses, (2) data collection (HEC Paris' Learning Center has many online data sources at your disposal), (3) testing of the said hypothesis/hypotheses with the collected data using statistical/econometric tests (such as linear regressions,

event studies, difference-in-differences regressions, etc.), (4) commenting at length about the **economic importance** of the findings (and not just focusing on the statistical significance or R^2 numbers), and (5) exposing the relevance of the findings for the readers in a larger context.

Put differently, as an example, a simple case study that replicates what was taught as a valuation exercise in one lecture of a previous class does **not** merit a letter grade of "A". Classes you are taking range from 12 to 27 hours of lectures that are worth 2 to 4.5 ECTS credit hours, respectively. Do **not** expect a portion of a class (worth 2 ECTS, for ex.) to generate an "A" that is worth 20 ECTS in the setting of the Research Paper. Put differently, less than 2 or 3 ECTS worth of an effort cannot lead to an "A" for the RP that is worth 20 ECTS.

Please note that the above-described set-up is put in place with clear objectives in mind:

First, it is there to allow you to be supervised by some of the best HEC Paris researchers and HEC Paris-linked industry captains. I am confident that you will make use of this opportunity and enjoy doing research under the guidance of some of the most brilliant people we have come to know, respect, and have the privilege to work with. Second, we very much hope that you will see the RP writing for the great opportunity it is to:

i. put your intellectual curiosity to work (and not just go through the motions to fulfil a cumbersome requirement),

ii. become experts in a particular area,

iii. learn new tools that you may deploy elsewhere later in your careers,

iv. produce a succinct and to the point piece of writing that reads well,

v. write a RP that will be graded/appreciated based on the quality of your output work (and not by weight of the print-out!)

We encourage you to talk to your classmates and form your pairs ASAP. If need be, we may have to combine you into pairs that are not your first-best. If the latter were to happen, please remind yourself that soon you'll be in professional environments in which you will be tasked to work in groups of no choosing of your own.

Profs. Evren ÖRS & Alina ROŞU

Academic Directors for the MIF, MiM's MIF, MD&F Programs
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More information **here**

Predicting the next Financial Crisis

In a context of growing global uncertainties, the prediction of the next financial crisis is a highly relevant topic. Students will conduct a meta-analysis of the existing financial crisis predictors, from macro-economic indicators and financial metrics to sophisticated statistical and machine learning forecasts. The predictors will be reproduced and compared using up to date, out-of-sample data, to distinguish overfitting from relevant indicators.

New Data for New Factors

The aim of the project is to discover new factors predicting asset prices. The students will develop systems to extract data from publicly available data sources, structured and unstructured, not yet strongly documented. They will then process this data to derive predictive factors that are not previously identified by the literature.

Financial Time Series Forecasting: from ARIMA to state-of-the-art Deep Learning Architectures

The students will do a systematic review of the literature about time-series predictions. They will then compare the efficiency of different methods of predictions on empirical examples, from the simplest autoregressive models to the most recent deep learning architectures (Transformers, LSTM, etc). The predictions will be used to construct trading strategies.

Feature Extraction & Embeddings for Financial Predictions

The students will explore various types of methods to generate predictive features from structured and unstructured data: Autoencoders, Image data extraction, Dimensionality reduction, Signal processing methods… The features extracted will be used in machine learning models to predict asset prices.

Datamining of Alpha Formulas

Following the literature on datamining and pattern discoveries, as well as the spirit of certain trading strategies of WorldQuant, the students will explore the various possibilities to automatically create and evaluate systematic, formula-based trading strategies. The final output of their project will be a combination of the generated strategies in a single meta trading strategy.

Supervisor: Bruno BIAIS

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How do Sovereign Wealth Funds delegate investment to asset managers?

Sovereign Wealth Funds (SWFs) are investment funds owned by a sovereign government. SWFs' assets under management amounted to more $12.9 trillion in 2024. In 2024, the top five SWF-host countries by AUM size were China (US $2.586 trillion), UAE (US $2.261 trillion), Norway (US $1.8935 trillion), Singapore (US $1.135 trillion) and Saudi Arabia (US $1.07 trillion) (https://globalswf.com/).
Empirical research, see, e.g., Cumming and Monteiro (2023), find that SWFs make large investments in alternative asset funds but have, other things equal, lower performance than other investors. One of the fundamental questions to be addressed by this memoir is whether such underperformance reflects agency problems.
SWF managers don't directly invest themselves in private equity, real estate, or infrastructure. Rather they delegate the management of their money to asset managers specializing in these asset classes. For example, at the end of 2023, the Norwegian sovereign wealth fund had 742 billion kroner (approximately 67 billion US dollars), managed externally by 103 asset management companies. Rose (2016) notes that "nearly all sovereign wealth funds make significant use of external managers, particularly for alternative asset classes that are difficult to access or demand labor intensive investment strategies, such as private equity, venture capital, and hedge fund strategies." This quote suggests that delegation is motivated by the need to rely on the skills of the managers, skills which the investor does not master. In fact, in the Invesco Global Sovereign Asset Management Study, respondents stated they often relied on external asset managers, and one of the respondents declared: "When… we have been forced to operate the business … that has historically not worked well, so we are moving away from that model and trying to leave all the management to general partners". As noted by Biais et al (2024), however, when delegation is motivated by superior understanding of assets by agents, information is likely to be asymmetric between the SWF and the asset managers.

This memoir will examine the process through which SWFs delegate the management of their money to asset managers, and how they manage this delegation. As shown in Biais et al (2024), information asymmetry between SWFs and asset managers creates an agency problem between SWFs and asset managers. In this principal-agent relationship, the former are the principals, while the latter are the agents. As shown in Biais et al (2024), to better align the incentives of agents with those of their principal, the latter should reward managers as a function of their performance, as well as allocate funds among managers as a function of their performance. Thus, when an asset manager performs better than another asset manager, other things equal, the SWF should reallocate assets from the latter to the former.

The goal of this memoir is to collect quantitative data and anecdotal evidence to assess whether such incentive schemes are implemented in practice. To pursue this goal, the students will collect data from SWFs and from international organizations, and from reports on SWFs and the asset management industry. With this data, the students will document the allocation of SWFs money across different types of funds, e.g., private equity, real estate, infrastructure, etc. The students will endeavour to use this data to test whether money is reallocated among asset managers as a function of their relative performance. The students will also endeavour to find data on managers' compensation and its relation to performance. This might be quantitative data, or statements, made in the reports of SWFs or industry reports.

Examples of data sources include the New Zealand Super Fund, which publishes a list of direct investments and external mandates, or the Ireland ISIF's reports, which provide information on portfolio and commitment schedules. The Australia Future Fund and the Norwegian Fund also provide information on their portfolios. These are just examples. One of the aspects of the students' work will be to search across sources, and integrate them, to build a useful dataset.

Interestingly, the above examples correspond to the SWFs of democratic countries. As discussed by the literature (see references below), the governance and strategies of SWFs also vary across countries in relation with countries' specific characteristics, such as corruption, transparency, or civil liberties. This offers another interesting avenue of empirical research for this memoir. For that research avenue, the students would have to collect data from a cross section of country, and examine if there is a link (across countries) between the characteristics of the SWF (types of investment, governance, relations with asset managers, portfolio dynamics) and the characteristics of the country (level of corruption, civil rights, development, transparency, etc.) Some of the latter data could be obtained from the Pew Research Center.

I would like to conclude this memoir topic with two friendly warnings:

- First, this is really a research topic, which means I don't know the answers to the questions, I don't know exactly how to frame the analysis, and I am not even sure I know exactly what I want to learn. That's what research is about: learning something new … that we don' know yet 😉
- Second, research is not about complication and technicality, it is about clarity and understanding. So your goal in this research memoir should be to work so hard that the questions, which initially seem complicated and unclear, in the end become very clear and compelling. This is a great thing to do, and a great experience to have 😊

References

Biais, B., H. Gersbach, J.C. Rochet, S. Villeneuve, and E.L. von Thadden, 2024, "Dynamic contracting with many agents", Working Paper, HEC.

Bortolotti, B., Fotak, V. and Megginson, W.L., 2015. “The sovereign wealth fund discount: Evidence from public equity investments”. *The Review of Financial Studies*, 28(11), 2993-3035.
Cuervo-Cazurra, A., Grosman, A., & Wood, G. T., 2023. “Cross-country variations in sovereign wealth funds’ transparency”. *Journal of International Business Policy*, 6(3), 306-329.
Cumming, D. and P. Monteiro, 2023, “Sovereign Wealth Funds Investment in venture capital private equity, and real assets funds’, *Journal of International Business Policy*, 16:1–26.
Megginson, W. L., Lopez, D., and Malik, A. I., 2021. “The rise of state-owned investors: sovereign wealth funds and public pension funds”. *Annual Review of Financial Economics* 13, pp. 247–270.
Pew Research Center, 2021. Public Trust in Government: 1958-2021. Pew Research Center. https://www.pewresearch.org/ Taylor, J. B., 2011.
Rose, P., 2016, “Sovereign Funds and External Asset Manager Fees: The Governance Connection,” Ohio State Public Law and Legal Theory Working Paper Series, No. 355

Supervisor: Richard BRUYERE

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More information **here**

1. **Cross-country performance/cost "puzzle" - to analyze the cross-country disparity in** structured **product fees**
    a. There is a strong variation in structured product costs by country in Europe. Hypothesis to be tested: structured products in some EU countries charge higher costs but do not deliver better projected outcomes. Cross-country variation highlights inefficiencies and consumer-protection / regulatory issues
    b. We would do cross-sectional regressions to assess the relationship between SRP costs and projected returns across different EU countries, controlling for factors such as payoff type, maturity, underlying asset class, + country effect (rate, household wealth, and other factors)
    c. Sources: for costs, payoff types, macro data – ESMA, SRP and ECB stats. On perf per country, to be crunched / analyzed (possible SRP reports)

2. **Performance of evergreen funds (esp-retail naitve) vs closed end funds:**
    a. Looking at different return/risk metrics and across asset classes and investment methods (secondaries, FoF, coinv, direct…). Is the retail investor getting a bad deal?
    b. Hypothesis tested: required liquidity provision for evergreen funds leads to a lower level of risk-taking for the same level of fees. We would estimate the market beta sensitivities on a number of fund parameters (Sharpe ratio, ...)
    c. Sources: Public fund fact sheets, Indefi in-house fund performance data base, online databases (Preqin, Pitchbook…)

3. **Impact of OCIOs / fiduciary management on pension / foundations / endowment investment performance**
    a. Looking at different return/risk metrics based on whether the investor is building their portfolio in-house or outsourcing. Is professional asset allocation worth it?
    b. Comparative statistical analysis of performance over different time periods, controlling for asset allocation discrepancies and other economic control variables
    c. Sources: Public investor performance data, internal data from leading endowments and foundations + OCIOs / fiduciary managers

4. **Performance of active ETFs relative to passive:**

    a. Looking at different return/risk metrics (incl. “tax alpha”) across US and European ETF products. Are active ETFs providing enough alpha relative to justify the higher fees?

    b. Modelling through a customized Fama-French model based on a number of parameters (asset class, portfolio market, management type, ...)

    c. Sources: Public fund data, accessed through SimFund (Indefi access)

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1. *Interaction between AI algorithms on financial markets:* an emerging literature uses computer simulations to understand how trading algorithms interact with each other, how their interaction affects the price discovery process, liquidity, or may lead to instability or other problems. Colliard, Foucault, and Lovo (2025) propose a "workhorse model" to study such questions. I would be happy to supervise research papers aiming at replicating the paper and extending it in any direction interesting to the students (are algos exposed to manipulation? can they trigger flash crashes? can they "talk" to each other? ...)

This topic requires some proficiency at programming in python or alternatives (e.g., Matlab). It is also recommended to follow my elective with Thierry Foucault on "Algorithms, Data, and Trading".

2. *Unraveling TACO: How does the government learn from market prices?:* When a government decision triggers a very adverse reaction from financial markets (e.g., government bond yields or stock prices), this may force the government to reverse its decision. In that sense, the government "learns" from market prices, and there have been prominent examples of this in 2025 (the famous "TACO trade" in the US, but also discussions on the French budget...). As noted in Bond and Goldstein (2015), market participants should anticipate that the government may reverse its decision, which should alleviate the market reaction in the first place. Little is known about the way this interaction plays out in practice. I would be happy to supervise research papers investigating how market prices moved around different government decisions and trying to test different hypotheses.

This topic requires conducting event study analyses, which any MIF student should be able to do. The main difficulty will consist in formulating clear economic hypotheses and interpreting the results, so a good understanding of financial economics is important (notions of informational efficiency in particular).

3. *What makes regulation complex?:* Colliard and Georg (2025) propose different tools to measure the complexity of regulatory texts, and make a variety of material available online (texts, dictionaries, experimental data, survey data, codes). I would be happy to supervise research papers aiming at extending this research in any direction of interest to the students. This could include comparing the explanatory power of different measures of complexity, extending the analyses to other texts or fields, perhaps even conducting other experiments. Another interesting direction could be to apply these measures to the complexity of financial contracts or prospectuses (structured products, hybrid bonds...).

This topic will likely involve conducting text analysis, which requires some familiarity with the associated tools (e.g., in python), or at least a willingness to learn them.

I am open to other research papers on topics that are of interest to me, in particular topics related to Finance and AI and financial regulation. If you want to suggest another topic, please send me an email describing the research question you want to investigate and the data you are planning to use or collect. Your proposal must include some analytical research (either analysis of a dataset, or numerical simulations, or a theoretical model). I do not supervise case studies.

Supervisor: Olivier DARMOUNI

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Generally, I am interested in topics related to financing the energy transition. Energy is central to the economy, with implications for growth, climate, competitiveness, inflation, etc. Ambitious social goals like de-carbonization or energy security can only be achieved with massive investment in energy infrastructure and power plants. Energy also matters for AI: the marginal cost of algorithms is electricity prices. However, these investments are very capital-intensive and need to be financed in some ways, by a combination of private funds (banks, private capital, stock market) and public funds (subsidies from taxes). There are many open questions on what drives investment and financing in the energy transition.

I am open to a wide array of RP on this broad topic, as long as the proposal contains (i) a clear economic hypothesis to be tested, (ii) good micro data, (iii) a statistical method for hypothesis testing (panel regressions, VAR, difference-in-difference, etc.).

Institutional knowledge is very important to understand questions related to energy. A few good sources for inspiration:

- General, World Energy Outlook: **https://www.iea.org/reports/world-energy-outlook-2024**, including Free data if you register
- A book on renewable energy finance specifically: "Renewable energy finance: theory and practice" by Raikar, Santosh; Adamson, Seabron, 2020. Link in HEC Library (read online):

Another great aspect of working on energy is that there is good micro data. Some examples:

- World Energy Outlook has free aggregate data if you register: https://www.iea.org/data-and-statistics/data-product/world-energy-outlook-2024-free-dataset
- The Global Energy Monitor: https://globalenergymonitor.org/, lots of great data on power plants and energy infrastructure worldwide.
- Cost of building different types of power plant in the U.S. from NREL: https://atb.nrel.gov/electricity/2024/financial_cases_&_methods
- Interconnection queues for U.S. energy projects in progress: https://emp.lbl.gov/queues
- And many other regional/regulatory sources

Some potential topics:

1. Why does the trajectory of energy transition look different across the world? Renewables, gas, and coal are growing at very different rates in different countries.
2. What are the effects of interest rates and the macro-financial cycle on energy investment? Shocks to central bank policies, credit markets, banks.
3. Did the 2022 energy crisis impact European energy investments and why?
4. How and why did the costs of different energy technologies change in recent years? Many technologies (like offshore wind) became significantly more expensive to build in recent years.
5. How fast are "brown" firms becoming green? For example, how has the business of European utilities changed over the past decade?
6. Which firms and investors own "brown assets"? See first part of my paper "Brown Capital (Re)Allocation" (https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4796331).
7. Are energy investments following datacenters investment?

Supervisor: François Derrien

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More information **here**

1) The evolution of the role of security analysts following Mifid 2

The Mifid2 European regulation affects the compensation brokers receive for providing research to their customers. How did it affect the organization of brokers (e.g., the number of security analysts they employ) and the role of security analysts (e.g., the number of firms they follow, their industry specialization,...) ?

2) The financing of growth firms in France / the EU vs. other areas of the world like the U.S. or Asia

The Draghi report argues that the lack of a financial ecosystem favorable to the financing of growth firms is one of the weaknesses of European economies. How does the European ecosystem compare with that of other areas of the world (e.g., the U.S. or China)? The students can consider for example the VC industry, its size, evolution and/or the equity market as a potential exit for early investors in new ventures.

3) The effects of the ESG backlash on prices, investors, firms, security analysts

How has the ESG backlash changed the environment of companies or other actors of financial markets in the U.S.? The students will examine how the ESG backlash has changed the incentives of one type of actor of financial markets (e.g., security analysts) and the real effects of this change.

4) The long-run performance of M&A transactions / IPOs / SEOs. Have the anomalies survived?

The academic literature has documented anomalies in long-run performance following corporate events like M&A, IPOs and SEOs. Did these anomalies persist in recent years? Why or why not?

5) Corporate Cash Holdings and Inflation

With recent periods of high inflation, firms' incentives to hold cash may change dramatically due to erosion of real value and higher opportunity costs.

6) Corporate payout policies and shareholder activism

Do activist interventions change firms' dividend or buyback policies?

7) ESG and firm value

In recent years, firms have been pushed by their various stakeholders to change their bahavior and "invest" in ESG. How has this affected their value?

8) Venture capital and IPO performance

Do venture-backed IPOs outperform non-venture-backed IPOs in the long run?

9) What are the determinants and consequences of issuing securities in crisis times ?

10) (In the same vein) Stock price reactions to SEO announcement in the recent crisis.

I am also happy to supervise any thesis that asks an interesting question broadly related to corporate finance or ESG (with a finance twist)

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Optimal hedging strategies using options

Supervisor: Matthias EFING

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1. **Data breaches and firm value**
   Explore the effect of data breaches on the cost of capital of firms or municipalities. Data on data breaches can be found at Privacy Rights Clearinghouse (PRC), a nonprofit organization that aims to promote data privacy awareness.

2. **Cash and Dividend Policy During the Pandemic**
   Explore the precautionary motive for cash during the pandemic. Investigate which firms cut dividends or repurchased shares, and which raised additional cash through debt or equity issuance. Use financial data to empirically assess these trends across sectors and firm sizes.

3. **Relative performance pay in executive compensation**
   Explore the extent to which firms condition the variable performance pay of their executives only on their own firm performance or, additionally, on the performance of the firm's closest peers/industry.

4. **Executive Compensation During the Pandemic**
   Analyze how performance-linked executive pay, particularly stock-based compensation, was impacted by the sharp decline in stock prices during the pandemic. Examine whether firms adjusted executive pay to compensate for these losses, and explore the empirical relationships between compensation changes and stock price performance.

5. **Gender Pay Gap** Study the evolution of the gender pay gap in recent years by examining compensation data across firms and industries. Explore how this gap has changed over time and whether policy interventions or increased awareness have impacted wage inequality between men and women.

6. **Scandals and Firm Value**
   Conduct event studies to analyze the stock market reaction to corporate scandals, focusing on whether firm value declines more severely in certain industries or scandal types. Empirically assess the effects on short-term and long-term performance.

7. **ESG Reporting Transparency**
   Investigate how government regulations have influenced corporate transparency in environmental, social, and governance (ESG) reporting. Analyze firms' responses to these regulations by examining changes in ESG disclosures before and after key policy implementations.

8. **2023 Banking Crisis**
   Examine which banking business models were most exposed during the 2023 banking crisis and how government interventions affected the stability of these banks. Analyze stock market reactions and explore how different banks fared based on their structure and risk exposure.

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More information **here**

1. **Nonbanks and Financial Integration in Europe**

A persistent concern for European policymakers is the lack of financial integration, which is often seen as a key obstacle to economic growth (see, e.g., the Draghi Report). When financial markets remain fragmented, capital fails to flow to its most productive uses. In particular, frictions in the banking sector are thought to limit cross-border capital flows within Europe. This thesis explores whether nonbank financial intermediaries help overcome these barriers. Specifically, it examines the degree of integration in nonbank lending markets by focusing on Collateralized Loan Obligations (CLOs). Using data on corporate loan holdings by CLOs, the thesis investigates whether these funds display a home-bias—overweighting loans of firms from their own country—or whether they allocate capital across Europe in a more integrated fashion.

2. **Industry expertise of CLO managers**

Collateralized Loan Obligations (CLOs) have grown into some of the largest institutional investors in corporate loans. Managed by large asset managers such as Blackstone, Apollo, and KKR, these funds actively trade loans within their portfolios rather than passively holding them. This thesis investigates the trading performance of CLO managers. In particular, it examines whether specialized expertise in certain industries enables CLO managers to generate abnormal returns through their trading activity.

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More information **here**

A/ Before selecting a project, please read the paper to which I refer to see whether you feel comfortable with the topic. All projects require manipulating and processing data.

B/ Before confirming the choice of a topic, please check data availability (I indicate sources for each project) and check with me whether the topic is still available.

C/ **You can propose alternative topics** as long as you have a clear question that you can address with available data and a clear benchmark (a scientific paper published in the Journal of Finance, the Review of Financial Studies, the Journal of Financial Economics, the Journal of Financial and Quantitative Analysis or the Review of Finance).

1. **Do machines beat professional forecasters?** Compare professional forecasters' forecasts (e.g., **macro-economic forecasts** by experts that can be obtained from central bank websites; e;g., FRED in the US) to those obtained with machine learning techniques (you can follow the approach here https://www.sciencedirect.com/science/article/pii/S0304405X24001338 but for different types of forecasters).
2. **Alternative data and earnings predictability.** Pick any alternative data of your choice (social media data, geolocation data etc.) and test whether it can be used (i) to forecast firms' earnings at short (less than 2 years year) and long horizons (more than 2 years) and (ii) stock returns over similar horizons. See for an example Green, Huang, Wen, Zhou (2019), "Crowdsourced employer reviews and stock returns," Journal of Financial Economics, 134, available here https://www.sciencedirect.com/science/article/pii/S0304405X19300662.
3. **Why do European firms list in the US?** Study the trend in European firms listing in the US, the characteristics of firms choosing to list in the US and the impact of the listing on their valuation and stock returns by conducting an event study. See https://www.ecb.europa.eu/press/fie/box/html/ecb.fiebox202406_07.en.html for background and https://tinyurl.com/mr3u6m58 for an academic study on the value of a US listing.
4. **Why do firms go public via SPACs** (Special Purpose Acquisition Company. Replicate and update "SPACs" by Gahng et al. (2021), available here: https://academic.oup.com/rfs/article/36/9/3463/7067751. Make sure you can access the relevant data before taking this topic.

5. **Are retail investors smart?** Replication of "The wisdom of the Robinhood Crowd", Welch (2022), Journal of Finance, available here:

https://onlinelibrary.wiley.com/doi/full/10.1111/jofi.13128. See published paper to see how to access the data.

6. **Does mutual funds ownership make stocks more fragile?** Replicate and update to more recent data Greenwood and Thesmar (2011), "Stock Price Fragility" available here: https://www.sciencedirect.com/science/article/abs/pii/S0304405X11001474. The goal is to compute an index of stock price fragility based on mutual funds ownership. Required data are available by accessing WRDS at the HEC library.

7. **Does institutional ownership increase or decrease firms' CSR (social responsibility) performance?** Replicate Institutional shareholders and corporate social responsibility available here: https://tinyurl.com/mrzt87ce. Check first whether you can access the MSCI ESG KLD database (KLD), which provide CSR scores. Alternatively you can use another available dataset providing CSR score.

Supervisor: Jean-Pierre FRANCOIS

Email address: jean-pierre.francois@engie.com

More information **here**

General Financial Markets/International Financial Markets/Commodities

1. Application of game theory to oil producing countries: forecasting market player behavior a new oil order.

2. Towards a methodology to measure the role of speculation in energy price dynamics.

3. Impact of Renewable Energy on traditional energy markets: how the integration of renewable energy sources affects the market dynamics, pricing and trading of traditional energy commodities like oil, gas or power?

4. EU carbon market: market efficiency, speculation, price fundamental factors.

5. Towards a methodology to assess the global impact of sanctions on fossil energy trade.

6. Impact of LNG Imports on European Gas Prices: how increased imports of liquefied natural gas (LNG) affect gas prices in Europe ?

7. The role of energy storage in balancing electricity supply and demand: how battery storage impacts renewable energy systems?

8. European natural gas market connectedness: a spillovers analysis.

9. Relationship between carbon market and energy markets: is it asymmetric?

10. Relationships between green indices and oil prices: how oil prices impact the development of the green transportation market?

11. Cross-border energy trade and market integration: what's the impact of cross-border electricity trading on market efficiency?

Supervisor: Johan HOMBERT

Email address: hombert@hec.fr

More information **here**

Mergers & Acquisitions: Empirical Research on the Question of Your Choice

Students are invited to design their own research question within the broad field of M&A, focusing on issues such as deal drivers, post-merger performance, acquirer or target characteristics, or market reactions. The thesis should involve working with real-world data, applying quantitative methods, and conducting rigorous empirical analysis. The aim is to provide original insights into the financial, strategic, or economic dimensions of M&A activity. Students are free to explore international settings and diverse datasets, provided their question is well-motivated and testable.

Supervisor: Jessica JEFFERS

Email address: jeffers@hec.fr

More information **here**

Generally speaking, I am interested in supervising theses in the following areas:

1. Labor mobility
2. Competition
3. Sustainable and impact finance, especially in alternative markets (e.g. private equity, debt) or regulation

Here are some examples:

Labor mobility

a. Do people who switch jobs more get promoted faster?
b. Where do we see the most enforcement of non-compete agreements (agreements that prevent employees from going to work for a competitor)?

Competition

a. Do connections among firm leaders (e.g. serving on the same boards) cause anticompetitive behavior?
b. Is there political favoritism in the approval of M&A deals?

Sustainable/impact finance

a. What kind of disclosure do companies provide in their impact reports, and who reads them?
b. What are the consequences of new European regulation like SFDR and MiFID II on investment funds, or CSRD or CBAM on corporations?

I am also happy to oversee paper replication/extension projects in any of these topics.

Supervisor: Ahmed KEBAIER

Email address: kebaier@math.univ-paris13.fr

More information **here**

Solar Energy Risks: Stochastic Modelling and Optimal Investment/Insurance Strategies

The accelerating integration of renewable energies into financial markets raises new challenges for risk assessment and optimal decision-making. In particular, solar energy production is inherently uncertain, driven by weather fluctuations, cloud coverage dynamics, and seasonal variability. These sources of randomness translate into non-trivial risk exposures for energy producers, investors, and insurers.

This thesis will focus on the development of stochastic models for solar radiation (e.g. diffusion processes with jumps, regime-switching models, or fractional approaches capturing long-range dependence) and their calibration to empirical radiation data. Building on this, the candidate will investigate the implications for risk management and valuation, such as:

- Pricing and hedging of weather derivatives linked to solar output,
- Optimal portfolio allocation strategies for investors exposed to solar production revenues,
- Design of insurance contracts mitigating production shortfall risks,
- Analysis of the impact of climate variability on long-term risk measures.

The work will combine advanced stochastic modelling (SDEs, L´evy processes, or rough volatilitystyle models) with applied tools from mathematical finance (risk-neutral vs. real-world measures, optimal control, and numerical approximation schemes).

Expected contributions include both theoretical insights (modelling frameworks, tractable dynamics) and practical applications (simulation studies, pricing/hedging algorithms, policy implications).

Supervisor : Augustin LANDIER

Email address: landier@hec.fr

More information **here**

1. Index Inclusion and Exclusion Effects

This study will investigate the market impact (price and volume) of stock additions to and deletions from major indices such as the S&P 500. The objective is to analyze how these inclusion and exclusion effects have evolved over time, focusing on both price reactions and trading dynamics. The project will also assess the profitability of investment strategies that seek to exploit stock-return predictability around index rebalancing events. Specifically, you will (i) simulate a strategy assuming perfect foresight of constituent changes *N* days in advance, and (ii) explore predictive signals or firm characteristics that could help anticipate forthcoming inclusions or exclusions.

2. The Predictive Power of Short Interest for Stock Returns

This project examines whether short interest contains information about future stock performance. You will analyze both the level and the change in short interest as predictors of subsequent returns, controlling for known risk factors such as size, value, and momentum. The analysis will explore how this relationship evolves over time and whether short interest signals are stronger in particular sectors or around earnings announcements. Finally, you will backtest a trading strategy designed to exploit these patterns.

3. Do Short Sellers Anticipate ESG Controversies?

This study explores whether short sellers anticipate upcoming ESG-related controversies before they become public. You will match short interest data with firm-level ESG news and controversy events to test whether short interest tends to rise ahead of negative ESG incidents. The analysis will distinguish between different types of controversies (environmental, social, and governance) and evaluate subsequent stock price reactions, thereby assessing the informational role of short sellers in ESG contexts.

4. Predicting Analyst Forecast Revisions

This project aims to model and predict analyst earnings forecast revisions using machine learning techniques. The first objective is to characterize individual analysts' behavior: when do they choose to update their forecasts, and can the direction of revisions be predicted? The analysis will investigate whether these revisions depend on peer analysts' actions or on market-level signals. Several modeling approaches—ranging from traditional econometric methods to modern machine learning algorithms—will be compared in terms of predictive accuracy and interpretability. Finally, the project will test whether the model's predicted revisions contain information about future stock returns, assessing their potential value for investment strategies.

Supervisor : Stefano LOVO

Email address: lovo@hec.fr

More information **here**

• ESG rating business model:
The rise of SRI (Socially Responsible Investing) has generated a sharp increase in information about the non-financial performance of firms. The amount spent by investors on purchasing ESG research and analytics has tripled between 2016 and 2021. Who pays for this information, the companies that are rated or the investors? What are the consequences for the **reliability** of these ratings?

The following thesis topics are proposed by the HEC Center for Impact Finance Research in collaboration with BNP Paribas

• Financial innovation:
What are the next frontiers in leveraging financial engineering to develop fit-for-purpose investment tools designed to address social and environmental issues?

• Impact investing:
What practical tools can help impact investing funds move from siloed and single-point approaches to solving societal challenges, toward more systemic ones?

• Retail investor preferences:
Why is investors' mindset not changing in line with facts and physical reality? Can behavioral sciences help shift the trillions needed to accelerate the transition to a net-zero, nature-positive, and more inclusive economy?

• Institutional investors:
Asset owners are institutions or individuals that ultimately own and bear the economic risk of investments (e.g., pension funds, sovereign wealth funds, insurance companies, endowments, and foundations). To what extent are asset owners playing a role in responsible investing and stewardship? Do they influence corporate behavior through their voting power and engagement policies?

Supervisor: Vincent MAURIN

Email address: maurin@hec.fr

More information **here**

1. **Estimation and asset pricing properties of the convenience yield for government bonds (US and other countries).** The thesis could replicate and update some previous work (e.g. https://www.journals.uchicago.edu/doi/10.1086/666526). Another related idea would be to estimate the risk-free rate, free of the convenience yield, similar to that work:

   https://www.sciencedirect.com/science/article/pii/S0304405X21002786

2. **What drives banks and other financial institutions' sensitivity to interest rate risk**? The 2023 banking crisis revealed the exposure of some banks to interest rate risk, as some banks failed when the Fed hiked rates. The purpose of the thesis would be to measure banks' sensitivity to that risk and the potential drivers of this sensitivity.

3. **Case study of Atos focusing on its debt policy, its financial difficulties and the renegotiation process.**

4. **The relationship between crypto prices and traditional asset factors. This paper could update the work by Liu & Tsyvinski**:

   https://academic.oup.com/rfs/article/34/6/2689/5912024?login=false.

5. **Market Neutrality and Market Efficiency of Central Bank Policies.** The thesis should explore the effects of asset purchase programs by the ECB on the cost of credit for affected firms. The focus should be on the concept of market neutrality and the recent tilts that have been introduced in the ECB framework. The topic needs to be refined, but a starting point can be found here: https://www.ecb.europa.eu/press/key/date/2021/html/ecb.sp210614~162bd7c253.en.html?

Supervisor: Nicolas NAILLON
Email address: nicolas.naillon@gmail.com

More information **here**

Nicolas Naillon, Financial Modelling Instructor, Managing Director Head of Healthcare Nomura

The topics below are examples of previous thesis and areas of interest. Ideally you should be making your own topic based on your own areas of interest. I am available should you have any questions on your thesis (Nicolas.naillon@gmail.com)

- Have M&A transactions in the recent past created value on the short term for the acquirer? An empirical study of market reactions of listed companies acquiring another company (vs premium announced and dilution announced)
- Have M&A transactions in the recent past created value on the long term? A study of long term benefit created by an acquisition or a merger (both empirical study and review of literature already published on the topic)
- Can we predict the "fair value" price in corporate finance? A review of various means and methodology to value the company vs prices offered during M&A transactions (once can help him(her)self in looking at public fairness opinions)
- Are corporate financial models useful to predict the development of a company? Can we get it right and does it matter?
- All topics related to the accuracy of financial modelling (operation models, LBO, merger models) towards the reality of what happens
- All topics relating to M&A and private equity
- All topics related to corporate finance and healthcare companies (hospitals, nursing homes, pharma, medtech, biotech): trends (a resilient market during the crisis, private equity ownership, M&A vs R&D investment: which choice and what consequences), premiums offered…
- Value creation through M&A: a clinical study on blockbuster deal evidence in the pharmaceutical industry
- All topics on corporate governance and networks in France and globally

Supervisor: Prof. Evren ÖRS

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More information **here**

Which Is a Better Hedge Against Inflation: Gold, Inflation-Indexed Bonds or Real Estate?

"Inflation worries fuel Japanese rush to buy gold" (Financial Times, 2023.09.06). "Investors pour billions of dollars into inflation-linked assets: US Tips bonds absorb record inflows while commodities and real estate regain allure" (FT, 2021.12.15). "Residential real estate is certainly a better inflation hedge ..." (FT, 2023.04.25).

This research paper will examine which of the above asset classes is a better hedge against inflation over the past. In a first step, the authors will read through The Economist, FT, etc. to find motivating arguments such as those above. Then they will search for and read academic research papers on the subject using www.ssrn.com and other similar websites. The latter will help them formulate hypotheses about inflation-hedge provided by the above listed asset classes. Then, they will collect data from websites such as Federal Reserve Bank of St. Louis's FRED database to test these hypotheses on a simple and unified regression framework (i.e., the same approach will have to be used for all the different asset classes under consideration).

Hedonic Pricing of Housing: An Application with the French Cadastral Data

Hedonic pricing approach relies on straightforward regression-based estimation methods to value homes whose values are linked with various characteristics that differ across properties (e.g., Sirmans et al, 2005, Journal of Real Estate Literature for a review). Through this research paper, the student pair will learn how to apply hedonic pricing methods, which are commonly used in the real estate investing and research, to publicly available French cadastral data. More precisely, this research paper will examine how to price real estate units (one of the following segments: apartments, houses, land lots) using data available on:

https://www.data.gouv.fr/en/datasets/demandes-de-valeurs-foncieres-geolocalisees/

Ideally, part of the sample will be used in the estimation of the regression model, the second part will be used to come up with predicted values based on the observable "hedonics" (estimated coefficients) and compare them with the actual values. This will allow testing the predictive power of the model.

Replication Papers: Stock Price Reactions to … using the Event Study Methodology

Many pairs/teams of students can work under this "family" of topics:

There is a long list of academic research papers on examining stock price reactions to various corporate announcements using the "Event Study" methodology (e.g., "Market Reaction to Corporate Press Releases" by Neuhierl et al., 2013, The Journal of Financial and Quantitative Analysis). Pairs of students can examine stock price reactions to one of the following events: dividend omission/cuts, dividend increases, share buy-backs, mergers and acquisitions for targets or acquirers, company level strike actions, CEO deaths, government or regulator announcements, etc. To do so, pairs of students will be asked to: (a) choose a particular type of event after discussing with me, (b) do a research on www.ssrn.com for the academic literature on stock price reactions to that particular type of event to form hypotheses, (c) sign-up to WRDS platform (through the HEC Paris Learning Center, i.e., HEC Paris Library), (d) read/learn about "Event Study Methodology" using on-line resources that I provide links to, (e) search for at least 100 firm-level public announcements for publicly traded US firms in news-search engines at different dates for the chosen event type, (f) collect US Stock Market data for the sample of firms thus obtained, (g) conduct the Event Study, (h) comment on the findings, (i) conclude.

SLOOS and US Banks' Lending

This research paper will examine the link, if any, between The Federal Reserve's Senior Loans Officers' Opinion Survey (SLOOS) and loans made by the US banks. To do so, the student pair will replicate parts (to be determined by discussing with the advisor) of the Morgan and Lown (published in the Journal of Money, Credit, and Banking, vol. 38, no. 6, September 2006, pp. 1575-1597) paper with more recent data to be obtained from the Federal Reserve Bank of St. Louis's FRED database as well as other Federal Reserve datasets from the Board of Governors' website. This research paper will allow the student-researcher-pair(s) to have a better understanding of how to use publicly available data to estimate simple empirical models to better understand the link between changes in banks' lending standards and business cycles.

Euro-Area Countries' Credit Standards Surveys and Macro Outcomes

Many pairs/teams of students can work under this "family" of topics:

This research paper will examine the link, if any, between European central banks' credit standards surveys (which are at the country-level) and loans made, and macroeconomic outcomes observed for the said countries in the following quarters. To do so, the student pair(s) will replicate parts (to be determined by discussing with the advisor) of the De Bondt et al. 2010 working paper ("The Euro Area Bank Lending Survey Matters: Empirical Evidence For Credit And Output Growth") with more recent country-level data to be

obtained from the ECB's online Bank Lending Survey Statistics. This research paper will allow the student-researcher-pair(s) to have a better understanding of how to use publicly available data to estimate simple empirical models to better understand the link between changes in banks' lending standards and business cycles.

Differences in the Euro-Area Banking Systems' "Deposit Betas"

This research paper will examine the differences in the responses of Euro-area banking systems' average deposit rates to the ECB's monetary policy changes. Please refer to:

https://libertystreeteconomics.newyorkfed.org/2023/04/deposit-betas-up-up-and-away/

The original research idea would be to calculate the deposit betas for different banking systems in the Euro-Area. This research paper would require that the student-researcher-pair collect data on deposit levels and deposit interest expenses for each country's banks on a quarterly level and compare the changes in the actual or "imputed deposit rate" (= deposit expenses/total interest-paying deposit balances) for different countries with the changes in the ECB's monetary policy rates. This would give the researchers an idea of the differences in the monetary policy rates' pass-through across various European countries.

As an alternative, the same exercise can be done with regional US banks located in different states of the USA.

Supervisor: Ciaran ROGERS

Email address: rogers@hec.fr

More information **here**

1. **Performance of Carry Trade Investments**

In recent years, investors have been increasingly taking on carry trades – a strategy of going short low-interest currencies and going long high interest rate currencies (for example, short Japanese Yen, long Australian dollar). This has become so prominent that, in recent months, when the trade suddenly performs badly, there is a large unwind, causing an unravelling of financial markets (see, for example, FT Article).

How profitable have these trades been in recent years? Do they generate any alpha? Is there a negative skew in returns, where it performs badly when there is an economic downturn? Have any of these characteristics changed over time, depending on the currency and/or region?

I can supervise several theses on this topic, as long as they use different data and/or answer a different specific question

2. **Monetary Policy and Medium-Term Inflation Expectations**

The mandate of many developed economies' central banks (European Central Bank, Federal Reserve, Bank of England) is to ensure inflation is close to 2% over the medium term. As a result, financial market expectations of inflation over the medium term is a key barometer for evaluating the success of central banks in achieving price stability. In the recent post-COVID high inflationary episode, however, expectations have risen significantly above 2%, and is becoming a very big concern for central banks. So what are they doing in response? In this project, you will investigate if central banks have changed their policy rules as a result. Have they become more response to inflation surprises than before to convey credibility? Are they more responsive because inflation is higher, or is it more because there is higher risk of low probability tail events of big inflation spikes? This is something you can test using widely available financial market data. The recent high-inflation episode is an important additional observation period to test this. Does the effect vary across countries, and if so why (past credibility of being close to inflation target pre-COVID)?

I can supervise several theses on this topic, as long as they use different country data, except for the US and for Europe.

3. Inflation Hedges: What Assets are Best?

In recent years, many countries have experienced unprecedented inflation post-COVID. In response, investors have rebalanced heavily to assets seen as traditional inflation hedges (gold, Inflation-Linked bonds, real estate, certain equities etc).

This project aims to understand which of the above asset classes is the best inflation hedge historically, and whether or not this relationship has changed in recent years. This will involve surveying recent reports on The Economist, the FT etc to get motivating arguments, followed up by reviewing some recent academic literature. This will help the students understand what regressions to run and how to formulate their argument. They can then get the data easily from Bloomberg, FRED etc.

I can supervise several theses on this topic, as long as they use different country data.

4. High Inflation US Household Asset Portfolios

In the paper by **Leombroni, Piazzesi, Rogers and Schneider (2022)**, we document how, over the period of high-inflation in the late 1970s, we see that a) wealth/GDP ratios of households fell considerably, and b) households re-allocated from housing to equity. Both of these phenomena can be explained by high inflation – households prefer to spend than save, and housing is seen as a better inflation hedge than equity.

In this project, students will firstly replicate the results of this paper. Secondly, it will then use the most modern data to understand whether or not households also rebalanced in a similar way as a consequence of the recent high inflation episode post-COVID in the US. Students can then dig deeper and understand whether or not certain households rebalanced more than others (wealthier/younger/equity participants etc).

5. Euro Area Sovereign Bond Yields

In the early 2010s, the Euro Area experienced a sovereign debt crisis. A combination of high government debt levels and a deep recession lead to large increases in government bond yields for Spain, Italy and Greece, amongst others. This worsened the recession because it necessitated austerity measures that were required to stabilize public finances, but at the cost of economic pain to households. This matters for the ECB as it induces strong deflationary pressures. In response, the ECB has conducted a variety of policies aimed at mitigating these effects by minimizing the movements of their government bond yields relative to safer issuers such as Germany. In this project, you will investigate if this was effective, and more importantly if it remains so. For example, does the Italian-German bond yield spread no longer respond to economic shocks? Does this relationship appear to change with ECB policies? Can you identify certain shocks (e.g. macro data announcements) that the spread is still sensitive to recently? This is an open question ripe for empirical analysis.

6. Risk Exposures of European Life Insurers

We want to understand the risk exposures of European life insurers to interest rate risk as well as sovereign risk in recent years. In other words, how sensitive are equity prices to a measure of interest rate risk and sovereign risk that you will construct as part of this project. Has the risk exposure changed over time, and if so, how? Is this becoming a key systemic risk in the system? What are the reasons behind the change in the risk.

The common consensus is that life insurers lose money when interest rates fall, but this has recently been turning on its head. We want to understand if this is true, and if so why.

This is building on existing work that I am already actively working on, and so will build on knowledge I have already accumulated over the past year.

Supervisor: Alina ROŞU

Email address: rosua@hec.fr

More information **here**

Research Paper Topics (in Empirical Asset Pricing and Asset Management)

1) **Fund trades, active share and profits.** Study the relationship between how much a fund trades and the profits it makes. Have a look at the paper by Pastor, Stambaugh, Taylor (2017) – The Journal of Finance, Do funds make more when they trade more? Replicate the study on US data. If you have data from other countries that would be even better to see if the relationship holds elsewhere. Also have a look at the Active Share concept (and related papers) on the website https://activeshare.nd.edu/. Study how the "activeness" of a fund relates to its performance.

2) **(Active) Mutual funds and their benchmarks – A performance study.** Mutual funds have an incentive to declare a benchmark (index) that makes their (benchmark-adjusted) returns look "rosier". You can check the paper of Sensoy (2009) - Performance Evaluation and Self-Designated Benchmark Indexes. Investigate to which extent you find data available for the composition of the indexes (check on Bloomberg, the Learning Center, WRDS, iShares ETFs, etc.). Replicate the study and see if the conclusion still stands. If you have the chance to use data from other countries that would be even better.

3) **Relevant factors in asset pricing studies**. Start from the course slide that describes the most common factors used in asset pricing. Replicate some of the studies that are made or find new data sets on which to test the models. Data are available in WRDS and on the website of Prof. Kenneth French from Dartmouth (a frequent co-author of Prof. Eugene Fama): https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

4) **Skewness and asset pricing tests**. Is skewness priced, do people prefer returns that exhibit positive skewness?

5) **Asset pricing and liquidity.** Is there any connection? Read the papers of Prof. Lubos Pastor and Prof. Sadka on liquidity, read about the liquidity measure of Prof. Amihud and think about liquidity and asset prices.

6) **Information and asset prices.** Read the papers of Prof. O'Hara on PIN (Probability of Informed Trading) and think about how information arrival changes asset prices and about tests that you could conduct.

7) **Hedge Funds**. How does beta with respect to the market differ in up markets and in down markets? Does beta increase in down markets? Do an analysis for each HF style. See what is happening if you run the regression with dummies as opposed to running two separate regressions. Are hedge fund betas stable over time? Do a study of fees of HFs. Is there a relationship between fees and performance? Data from Credit Suisse (need to make an account with them) or HF Research (potentially available in the WRDS database – contact the Learning Center to get access). Read the book by Sebastian Mallaby – More Money than God: Hedge Funds and the making of a new elite

8) **Any topic** concerning mutual funds, hedge funds, asset management, AI and stock picking, asset pricing, investment by households (household finance).

Supervisor: Ioanid ROŞU

Email address: rosu@hec.fr

More information **here**

1. Estimation of risk and return at long horizons. Are stocks really less risky than bonds in the long run?

2. New risk factors (investment, profitability, low volatility, betting against beta, quality) versus old risk factors (size, value, momentum).

3. Quantitative asset management: Optimal portfolios with alternative assets, various constraints, transaction costs, and accounting for cash inflows and outflows.

4. Factor investing ("Smart Beta") in asset management. Example: How a pension fund should invest when considering exposure to various factors, such as inflation, exchange rate risk, or economic growth.

5. Machine learning and applications to: (i) hedge fund strategies and return predictability, (ii) banking and credit scoring, or (iii) text interpretation in finance.

6. The Black-Litterman model, or how to invest when the manager has views about returns. What if the views are not about returns but about volatility, correlation, or other variables?

7. How good is the performance of leveraged ETFs? Are they a good investment for the long run?

8. Performance of Hedge Funds and option-like strategies. Do Hedge Funds have positive alpha and zero beta?

9. The growth rates of companies and learning about profitability. How do we estimate the growth rates of companies and how do they evolve over time?

10. The equity risk premium and its connection with macroeconomics announcements.

11. Any other topic agreed with the student involving asset pricing, asset management, or mathematical finance.

Supervisor: Daniel SCHMIDT

Email address: schmidt@hec.fr

More information **here**

Note: topics are given on a first-come-first-serve basis. I am also open to your own topic suggestions.

1) **Climate Finance**. Global warming poses an existential threat to our economic well-being, health, and even survival as a species. After surveying the literature, the student will focus on one aspect of how global warming is affecting finance. For example, how do investors respond to the increasing negative publicity on "dirty" firms? Do stock returns reflect climate risks in a timely and adequate manner? How well do companies and investors respond to changing prices of carbon emission certificates? How much do investors and/or customers react to ESG scandals? Are green bonds priced differently compared to normal bonds?

2) **Hedge Fund Activism**. Hedge Fund Activists target firms that are perceived inefficient or undervalued with the aim of changing/influencing corporate policy. How successful are such activism campaigns? Can one predict which activist campaigns are more likely to succeed? Are they followed by long-term abnormal returns? Do they achieve corporate change?

3) **Topics in Behavioral Finance**. Behavioral Finance studies how cognitive biases of investors or managers affect financial decisions. After surveying the literature (excellent literature surveys have been written by David Hirshleifer (UCI) and Nicholas Barberis (Yale)), students can choose to work on a particular subquestion in this large field. It may be particularly interesting to explore a behavioural topic in less-studied markets such as real estate or betting markets.

4) **Efficiency of Betting markets**. Betting markets become increasingly popular as a way of betting on events in the realm of politics, economic policy, geopolitics, culture, and tech. For example, on https://polymarket.com one can download the history of betting odds, which allow to track the evolution of probabilities for various events. How efficient are these markets? Do they reflect new information in a timely manner? Are there arbitrage opportunities, i.e., instances whereby one can make a sure profit by trading on multiple offsetting bets simultaneously? Can the implied probabilities be used to predict the returns of other financial assets?

5) **Market Efficiency and Limited Attention**. According to the Efficient Market Hypothesis (EMH), stocks prices should reflect new information instantaneously. Yet, the literature has uncovered instances of both underreaction and overreaction to news. The post-earnings announcement drift—the tendency for stock prices to drift in the direction of the earnings news—is arguably the best-known departure from the EMH and has often been explained as a result of limited attention by market participants. Students will summarize the literature in this field and conduct a related empirical analysis. For instance, students could test whether investors respond to earnings announcements even though the announced earnings were already anticipated by the market. Another idea would be to test whether investors perhaps overreact to one-time profit effects (extra-ordinary income) by not realizing that this profit level is probably not sustainable. [Interesting data on revealed attention choices by people can be gleaned from: https://www.google.com/trends/]

6) **Lessons from Financial History**. Past centuries have seen many financial market bubbles and subsequent crashes. From the British South Sea Bubble in 1720 to the Dotcom Bubble in 2000, they all offer to teach us something about financial markets today. Students will choose a particular historical example, and relate and compare it to recent financial developments. [Potentially useful historical data can be found at: http://som.yale.edu/faculty-research/our-centers-initiatives/international-center-finance/data/historical-financial]

7) **Market Liquidity**. Market liquidity describes the ease with which investors can trade in financial markets. Sudden liquidity dry-ups in certain asset markets during the recent financial crisis have brought liquidity concerns to the forefront of academic research. After surveying the literature (an excellent survey has been written by Lasse Pedersen (CBS) and Yakov Amihud (NYU)), students will focus on one particular aspect in this field.

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Test of Market Efficiency

Unbeknown to many market participants, central banks signal their trading activity as their mandates contain the objective of price stability. We will study the impact of the announcements of future JGB trades of the Bank of Japan. The null hypothesis is that the markets are perfectly efficient, and that there should not be systematic gains for portfolio managers. This research project aims to expose students to the analysis of material financial information in the text format as the central banks communicate on their actions.

We will employ AI techniques with NLP models such as the Transformer based BERT Language Model fine-tuned on the Stanford Question Answering Dataset. Stagnol, L. et al. (2023). Answering clean tech questions with large language models. Available at SSRN: https://ssrn.com/abstract=4663447 is an illustration of the performance of these models.

Inflation here, inflation there

The return of inflation has brought forward literature on cost pass-through such as Lafrogne-Joussier, R. (2023). Focus 94: Inflation en 2023 — Focus Cost Pass-Through and the Rise of Inflation. Conseil d'Analyse Économique. https://cae-eco.fr/focus-94-inflation-en-230509.

We will employ NLP techniques to identify the pricing power of superstar firms through their "pass-through" claims in the current tariff environment. We will run an econometric test of the obtained refined pricing power through the impact of increase in cost on the firms' margins.

This research project aims to expose students to the capacity of Transformer based Language Models to be fine-tuned to specific research questions. Students can refer to Stammbach, D., et al. (2022). Environmental claim detection. SSRN. https://ssrn.com/abstract=4207369 for an example of fine-tuning.

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Topics for HEC Master Theses in Finance

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HEC Paris
Associate Professor of Economics and Entrepreneurship
Department of Economics and Decision Sciences
Scientific Director, ION Management Science Lab – Paris

Introductory note for students.

This portfolio offers six projects in the topics economics of mentorship and entrepreneurship, leveraging data from the Creative Destruction Lab Paris. You will work with real meeting recordings, venture descriptions, mentor profiles, and the sequence of objectives (founder draft → Objective Setter proposal → Objective Critiquer version) to study how conversations reshape roadmaps—covering founder misdiagnosis and "resets," engagement and information sharing, expertise fit, conversational patterns that trigger changes, and the clarity and actionability of final objectives.

Skills note

All these six projects require comfort with Python or spreadsheets, cleaning and merging small datasets descriptive statistics, and clear, reproducible write-ups (notebooks and tidy folders). In addition, they require advanced programming or generative-AI techniques (e.g., dialogue feature extraction, semantic similarity modeling, prompt design and evaluation).

Academic integrity

Across these projects, I expect students to make extensive, responsible use of large language models to accelerate skill acquisition (code explanations, debugging, drafting rubrics and documentation), while keeping analyses transparent and reproducible. I also expect responsible use of large language models in the writing of the thesis.

Topic 1. Economics of Mentorship and Entrepreneurship (Creative Destruction Lab Paris)

Context for students.
Creative Destruction Lab Paris is a mentorship-based accelerator for science and technology ventures. Founders join to sharpen strategy, commit to concrete near-term objectives, and access experienced mentors who are entrepreneurs, operators, investors, and scientists. Mentors help by stress-testing ideas, proposing specific next steps, and opening doors to people and resources.

Before each meeting, founders submit a short description of their venture and a draft list of objectives for the next period. During the small group meeting, several mentors meet the founders together. One mentor is designated as the Objective Setter; this person proposes a clear, prioritized set of objectives

for the founders. A second mentor is designated as the Objective Critiquer; this person reviews the proposal on the spot and suggests clarifications or changes to improve feasibility, evidence alignment, and focus. The small group format brings more than one mentor to the table to surface different perspectives and reduce individual bias. Your projects use: (i) audio recordings and transcripts of these meetings, (ii) the venture descriptions, (iii) mentor profiles, (iv) who served as Objective Setter and Objective Critiquer, and (v) the three versions of the objectives (founder draft, Objective Setter proposal, and Objective Critiquer modifications).

Project 1. From Founder Draft to Final Objectives: Who Changes What?
In this project, you will compare the founder's draft objectives with the Objective Setter's proposal and the Objective Critiquer's modifications. You will build a simple taxonomy of edits (for example: additions, removals, re-ordering, changes to measurability, deadlines, and ownership). You will quantify how much the content shifts at each step using straightforward text-comparison tools, and you will relate larger or smaller shifts to mentor profiles and to elements visible in the transcript (for example, clarifying questions or references to data, customers, or pilots). The deliverable is a clear description of typical change patterns and the conditions under which founders' plans are most reshaped. **Basic knowledge required**: Comfort with Python and spreadsheets, simple text comparison and regular expressions, introductory natural language processing for short documents, and basic statistics

Project 2. Conversation Patterns that Precede Objective Changes
Using the recordings and transcripts, you will trace when new elements first appear in the objectives and what was being said just before those changes. You will mark moments such as probing questions, constructive challenges, and references to evidence, and link them to later edits by the Objective Setter or the Objective Critiquer. The aim is to describe common sequences—what types of questions or prompts tend to lead to adoption, rejection, or refinement of an idea—and provide illustrative excerpts that show how the conversation moved the objectives. **Basic knowledge required:** Python for handling time-stamped transcripts, introductory natural language processing for speaker turns and dialogue tags, and basic probability/statistics (optional: introductory time-to-event analysis)

Project 3. Mentor–Venture Expertise Fit and the Shape of the Final Objectives
You will summarize each mentor's expertise from their profile and each venture's needs from its description, then create a simple measure of how well they match. You will examine whether a closer match for the Objective Setter or for the Objective Critiquer is associated with a greater share of the final objective reflecting that person's contributions, and whether differences in match are linked to the type of objectives that emerge (for example, operational steps versus strategic positioning, short-term tasks versus longer-term direction). The output informs how assigning roles to mentors with different backgrounds can change the final plan. **Basic knowledge required**: Python for text processing, introductory natural language processing for semantic similarity, and introductory econometrics (multiple regression with controls).

Project 4. Scoring the Quality and Actionability of Objectives
You will design a brief rubric to score objective quality along a few dimensions such as specificity, measurability, clear timing, named owner, and link to stated risks or assumptions. You will apply the rubric to the founder draft, the Objective Setter proposal, and the Objective Critiquer version, and then study which conversational behaviors are most often associated with higher-scoring final objectives (for example, the number of clarifying questions or insistence on evidence). The result is a practical scoring guide plus a short list of mentor practices that reliably improve clarity and actionability. **Basic knowledge required**: Python or spreadsheets for scoring and aggregation, introductory natural language processing for rubric automation, and basic measurement concepts (inter-rater agreement and simple reliability checks).

Project 5. When Do Critiques Change the Plan? A Simple Proposal–Review Model and Test
You will develop a short, intuitive model of how a proposal from the Objective Setter is reviewed by the Objective Critiquer, and when it is endorsed versus modified. You will translate the model's main ideas into measurable conditions using the data (for example, how different the two mentors' expertise is, how much relevant evidence was put on the table during the discussion, and how confident the Objective Setter sounded), and then test the model's predictions against the observed frequency and size of modifications. The project produces a concise theory note and an empirical summary that together offer guidelines for when critique adds the most value in small group meetings.
Basic knowledge required: Python for text processing, introductory natural language processing for semantic similarity, and introductory econometrics (multiple regression with controls).

Project 6. Founder Misdiagnosis and Objective Resets in Small Group Meetings
In this project, you will study how founders' pre-meeting objectives are corrected during the small group meeting. Using the founder's draft objectives, the Objective Setter's proposal, the Objective Critiquer's modifications, mentor profiles, venture descriptions, and the meeting transcripts, you will (i) define a simple rubric for "misdiagnosis" (for example: over-engineering reliability targets, planning organization design too early, spreading across too many customers or markets, underweighting cash runway, or skipping a near-term proof-point); (ii) detect "resets" where mentors redirect the plan toward a more informationally efficient next step (for example: reduce reliability targets to what is sufficient now, narrow the go-to-market, prioritize cash runway, commit to one tangible proof-point); and (iii) identify conversational triggers that precede the reset (probing questions, requests for evidence, reframing). You will describe how often and how strongly plans are reset, illustrate the typical sequences with short transcript excerpts, and relate resets to mentor–venture expertise fit and to the balance of voices in the room (for example, whether early strong statements appear to anchor the discussion or whether diverse perspectives are heard before a decision).
Basic knowledge required: Comfort with Python and spreadsheets, simple text comparison and tagging for short documents, introductory natural language processing for dialogue features, and basic statistics.

Project 7. Engagement, Information Sharing, and How Objectives Take Shape
In this project, you will measure how actively founders and mentors engage with one another during small group meetings and examine how that engagement influences what founders reveal, what mentors learn, how the conversation moves across topics, and how the objectives change from the founder's draft to the Objective Setter's proposal and the Objective Critiquer's version. Using speaker-attributed transcripts, you will construct simple engagement measures (for example: balance of speaking time, alternation of turns, response latency, number of genuine questions, depth of follow-ups in a single line of inquiry, and supportive back-channels such as brief acknowledgements). You will code concrete information shared by founders (for example: specific customer counts, reliability figures, revenue or runway numbers, experiment results, and named risks) and visible signs of mentor learning (for example: restatements to check understanding, explicit updates or corrections, and references to newly learned facts later in the meeting). You will relate these measures to the flow of topics (time spent per topic and the order in which topics appear) and to the content and quality of the final objectives (specificity, measurability, timing, ownership, and scope changes). The goal is to quantify when a more engaged conversation leads to richer disclosure, clearer mentor understanding, and better-formed objectives.
Basic knowledge required: Comfort with Python and spreadsheets, introductory natural language processing for dialogue analysis on speaker-attributed transcripts, and basic statistics or simple regression.

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- **Takeover and Activism Defense Mechanisms**

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- **The performance of private equity**. How does private equity perform as an asset class? What are the main issues when measuring the performance of private equity deals?

- **The valuation of intangible assets**. How do markets value assets which often do not show up in balance sheets? Are capitalization methods able to bride valuation gaps?

- **Dual-class shares**. Are there differences in the performance of firms dual-class shares? Do dual-class shares create or destroy value?

- **Corporate taxes**. How much taxes do companies really pay? How to reconcile the gaps between marginal and effective tax rates?

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Constructing optimal investment portfolios

Background readings:

- "Pricing currency risk" by Chernov, Dahlquist, and Lochstoer (Journal of Finance, 2023)
- "Pricing and constructing international government bond portfolios" by Randl, Simion, and Zechner (Working Paper)
- Useful data: https://mpelger.people.stanford.edu/data-and-code

Monetary policy and asset markets. Impact of macroeconomic announcements on asset prices

Background readings:

- "Monetary policy communication, policy slope, and the stock market" by Neuhierl and Weber (Journal of Monetary Economics, 2019)
- "The pre-FOMC announcement drift" by Lucca and Moench (Journal of Finance, 2014)
- "Asset pricing: a tale of two days" by Savor and Wilson (Journal of Financial Economics, 2014)
- "Exchange rates and monetary policy uncertainty" by Mueller, Tahbaz-Salehi, and Vedolin (Journal of Finance, 2017)

Government debt sustainability

Background readings:

- The behavior of U.S. public debt and deficits by Henning Bohn (Quarterly Journal of Economics, 1988)

Frictions and asset prices

Background readings:

- "Betting against beta" by Frazzini and Pedersen (Journal of Financial Economics, 2014)
- "Financial Market Dislocations" by Pasquariello (Review of Financial Studies 2014)

Macroeconomy and asset prices

Background readings:

- "Consumption fluctuations and expected returns" by Victoria Atanasov, Stig Moller, and Richard Priestley (Journal of Finance, 2020)
- "Good volatility, bad volatility and the cross section of stock returns" by T. Bollerslev, S. Li, B. Zhao (Journal of Financial and Quantitative Analysis, 2019)

- “Good and bad uncertainty: macroeconomic and financial market implications” by Segal, Shaliastovich, and Yaron (Journal of Financial Economics, 2015)
- Importance of market-wide information for explaining stocks’ returns (“R2”, Richard Roll, 1988, Journal of Finance)
- What explains the volatility of asset prices (Roll, 1984, “|Orange Juice and Weather”, 1984)

Investments and asset management

Background readings:

- “The level, slope and curve factor model for stocks” by Clarke (Journal of Financial Economics, 2020)
- “Factor timing” by Haddad, Kozak, Santosh (Review of Financial Studies, 2020)
- Volatility-managed portfolios by Moreira and Muir (Journal of Finance, 2017)

Any topic of your choice related to financial markets

Expectations hypothesis in fixed-income markets around the globe (Campbell and Shiller, 1991, "Yield Spreads and Interest Rate Movements: A Bird's Eye View", Review of Economic Studies)